

PROJECT DOCUMENTATION

Import Data Using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-8263

1. Introduction

Project Title: Import Data Using Transform Maps (Spreadsheet)

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This document describes the functional setup, architecture, operation, testing and maintenance considerations for the ServiceNow-based employee-data import solution.

2. Project Overview

Purpose

The solution imports employee information from a spreadsheet into ServiceNow, transforms the staged data into the Employee Test table, uses Employee ID for record matching, and presents the processed information through reports and an Employee Analytics Dashboard.

Features

- Spreadsheet upload through ServiceNow Import Sets.
- Employee Import staging table for source data.
- Sample Spreadsheet Import Transform Map.
- Five source-to-target field mappings.
- Employee ID coalesce for existing-record matching.
- Employee List, Employees by Location and Employees by Department reports.
- Employee Analytics Dashboard.
- Role and ACL-based access control.

3. Architecture

Frontend / User Interface

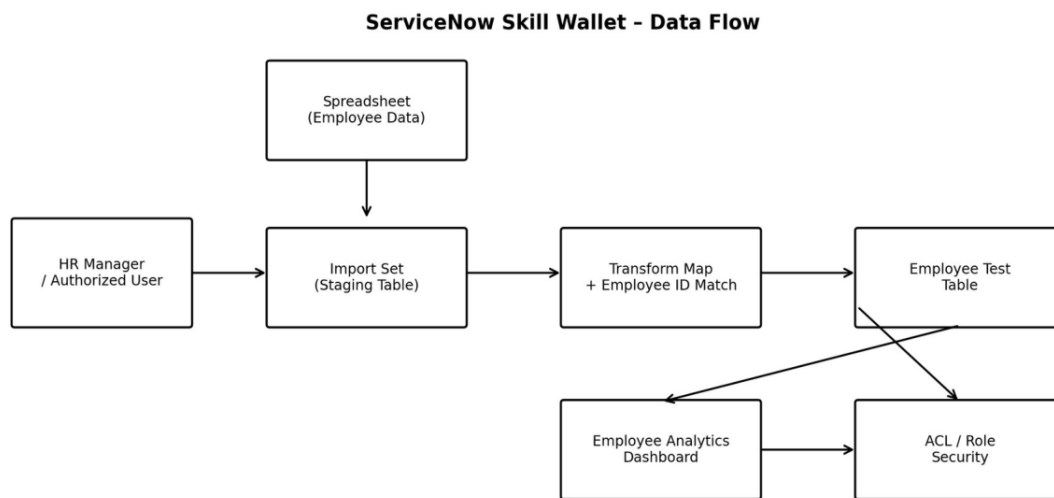
The user interface is the ServiceNow Web UI. HR or authorized users interact with Import Sets, transformation configuration, employee records, reports and the Employee Analytics Dashboard.

Backend / Application Logic

ServiceNow Import Sets handle spreadsheet ingestion and staging. The Sample Spreadsheet Import Transform Map performs field mapping and transformation. Employee ID is used as the coalesce field for matching existing records.

Database

The Employee Import table acts as the staging data store and the Employee Test table stores the transformed employee records. The platform database is managed by ServiceNow.



Architecture diagram.

4. Setup Instructions

Prerequisites

- Access to a ServiceNow Personal Developer Instance or equivalent ServiceNow environment.
- Appropriate administrative/configuration permissions for creating tables, Import Sets, Transform Maps, reports and ACLs.
- An employee spreadsheet containing the required columns.
- The required ServiceNow application/table configuration.

Installation / Configuration

1. Create or open the Employee Test table and define the required employee fields.
2. Create the Employee Import staging table and load the spreadsheet.
3. Create the Sample Spreadsheet Import Transform Map.
4. Configure the five source-to-target field mappings.
5. Set Employee ID as the coalesce field for record matching.

6. Run the transform and verify the transformation history.
7. Create the Employee List, Employees by Location and Employees by Department reports.
8. Add the reports to the Employee Analytics Dashboard.
9. Configure the required roles and ACLs for controlled access.

5. Folder Structure

Because the project is implemented inside ServiceNow rather than as a local MERN application, the logical project structure is organized by platform components:

Component	Purpose
Employee Test table (u_employee_test)	Target table containing transformed employee records.
Employee Import table (u_employee_import)	Staging table receiving spreadsheet records.
Sample Spreadsheet Import	Transform Map used for source-to-target mapping and transformation.
Reports	Employee List, Location and Department reports.
Employee Analytics Dashboard	Consolidated reporting view.
Roles & ACLs	Access-control configuration for employee information.

6. Running the Application

No local frontend/backend start commands are required. The solution runs in the ServiceNow instance.

1. Sign in to the authorized ServiceNow instance.
2. Open the Import Set / employee-data import functionality.
3. Load the employee spreadsheet into the Employee Import staging table.
4. Run the Sample Spreadsheet Import Transform Map.
5. Review transformation results and verify Employee Test records.
6. Open the Employee Analytics Dashboard to view the reports.

7. API Documentation

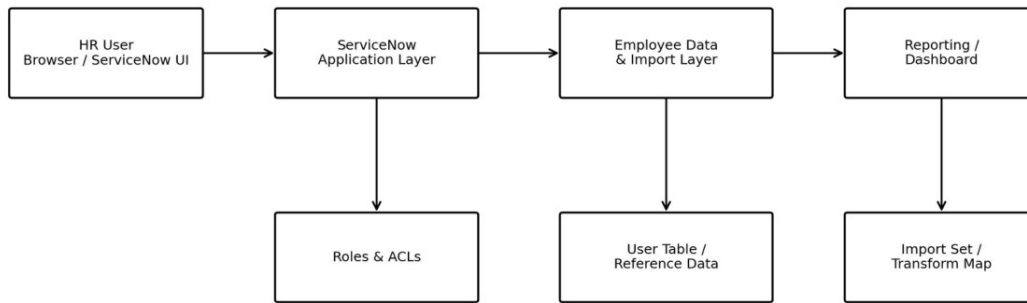
No external REST or third-party API is used by the implemented solution. Data movement is handled by ServiceNow Import Sets and Transform Maps within the platform.

8. Authentication

Access is provided through the ServiceNow instance's authentication mechanism. Authorization for employee-data operations is controlled through ServiceNow roles and ACLs. The project configuration includes HR-oriented access controls so that employee information is available only to authorized users.

9. User Interface

ServiceNow Skill Wallet - Technical Architecture



Field mapping interface.

The screenshot shows the ServiceNow Field mapping interface for a "Table Transform Map - Sample Spreadsheet Import". The interface includes a header with the ServiceNow logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. Below the header, there are buttons for "Copy", "Update", and "Delete".

The main section displays the field mapping configuration for the "Sample Spreadsheet Import" table. It includes the following fields and options:

- Name:** Sample Spreadsheet Import
- Source table:** Employee Import [u_employee_import]
- Active:** ☒
- Run business rules:** ☒
- Enforce mandatory fields:** No
- Copy empty fields:** ☐
- Create new record on empty coalesce fields:** ☐
- Application:** Global
- Created:** 2026-09-28 02:47:16
- Target table:** Employee Test [u_employee_test]
- Order:** 100
- Run script:** ☐

Below the configuration fields, there are buttons for "Copy", "Update", and "Delete". A "Related Links" section provides links to "Auto Map Matching Fields", "Mapping Assist", "Validate Coalesce Fields", "Transform", "Index Coalesce Fields", and "Run Point Scan". At the bottom, there are tabs for "Field Maps (4)" and "Transform Scripts".

Successful transformation result.

Field Maps (5) Transform Scripts			
⋮ ⚙️ — Actions on selected rows... New			
☐	Source field	Target field	Coalesce
	u_name	u_employee_name	false
	u_email	u_email	false
	u_department	u_department	false
	u_employee_id	u_employee_id	false
	u_location	u_location	false
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Employee Analytics Dashboard.

10. Testing

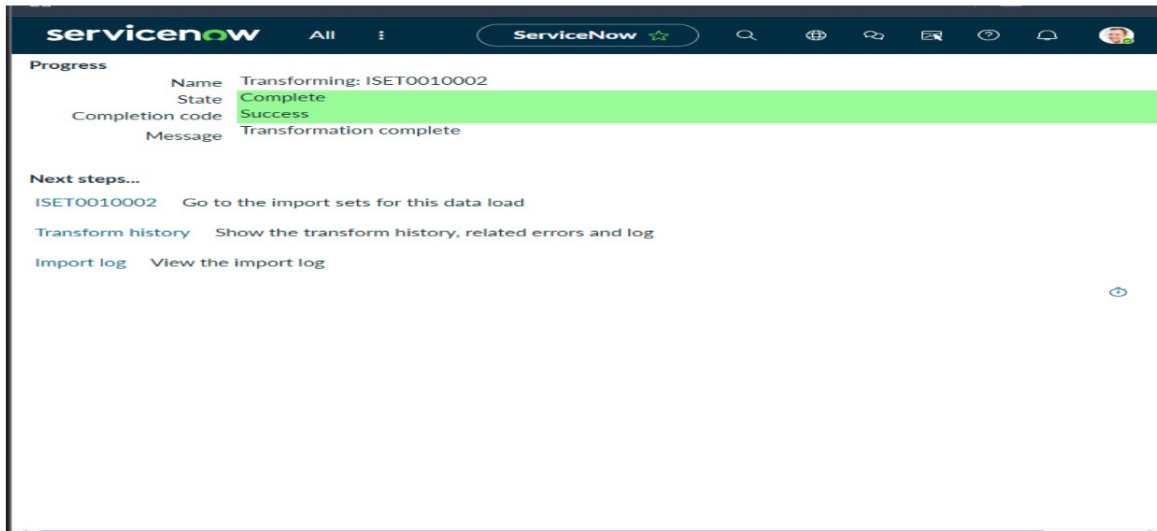
Testing covered functional field validation, spreadsheet staging, Transform Map mapping, Employee ID coalesce, bulk import, identical re-import, reports and the dashboard.

Test Area	Result
Data Import & Transform	2/2 test cases passed.
Coalesce / Deduplication	2/2 test cases passed.
Reports	3/3 test cases passed.
Dashboard	1/1 test case passed.
Total UAT	8/8 test cases passed.

Three bugs were identified and closed: duplicate records during re-import, blank Employee Name values, and an incorrect Employees by Location chart type.

11. Screenshots or Demo

The working demonstration is available in the ServiceNow Personal Developer Instance used for the project. The screenshots in this document provide evidence of the configured Transform Map, successful transformation and dashboard output.



Employees by Department report.

12. Known Issues

The solution depends on the source spreadsheet following the expected field structure. If source column names or data formats change, the Import Set and Transform Map configuration may need to be updated. The solution also requires appropriate ServiceNow permissions and a configured instance.

13. Future Enhancements

- Schedule recurring spreadsheet imports.
- Add stronger spreadsheet validation and error reporting.
- Expand the employee and training data model.
- Add dashboard filters and additional analytics.
- Introduce notification workflows for important data changes.
- Expand role-based access for different HR responsibilities.
- Add audit-oriented reports for import and record-change tracking.